

E9 205 Machine Learning for Signal Processing

Neural Networks - Generalization

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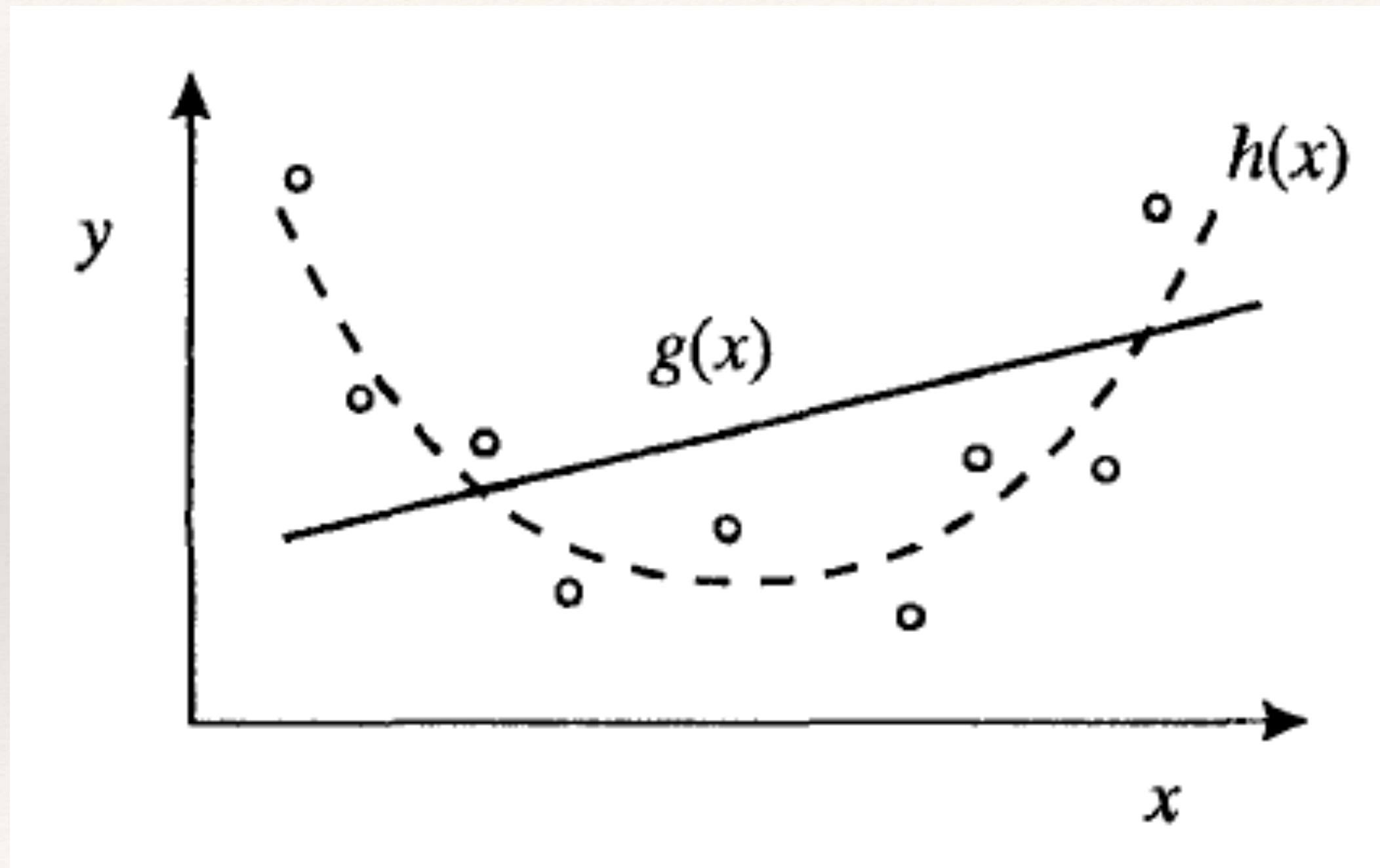


Bias and Variance In Neural Network Training

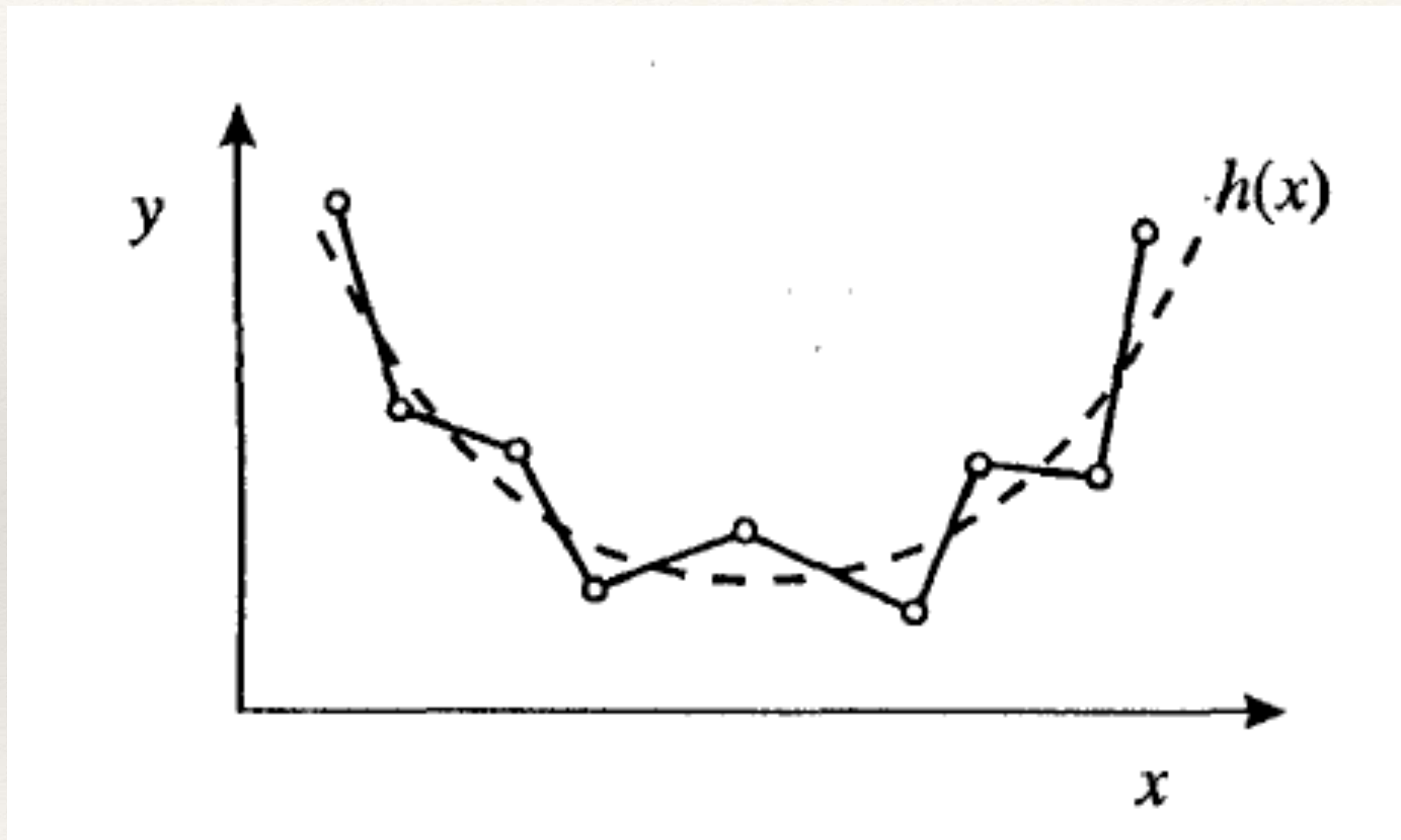
$$(\text{bias})^2 = \frac{1}{2} \int \{\mathcal{E}_D[y(\mathbf{x})] - \langle t|\mathbf{x} \rangle\}^2 p(\mathbf{x}) d\mathbf{x}$$

$$\text{variance} = \frac{1}{2} \int \mathcal{E}_D[\{y(\mathbf{x}) - \mathcal{E}_D[y(\mathbf{x})]\}^2] p(\mathbf{x}) d\mathbf{x}.$$

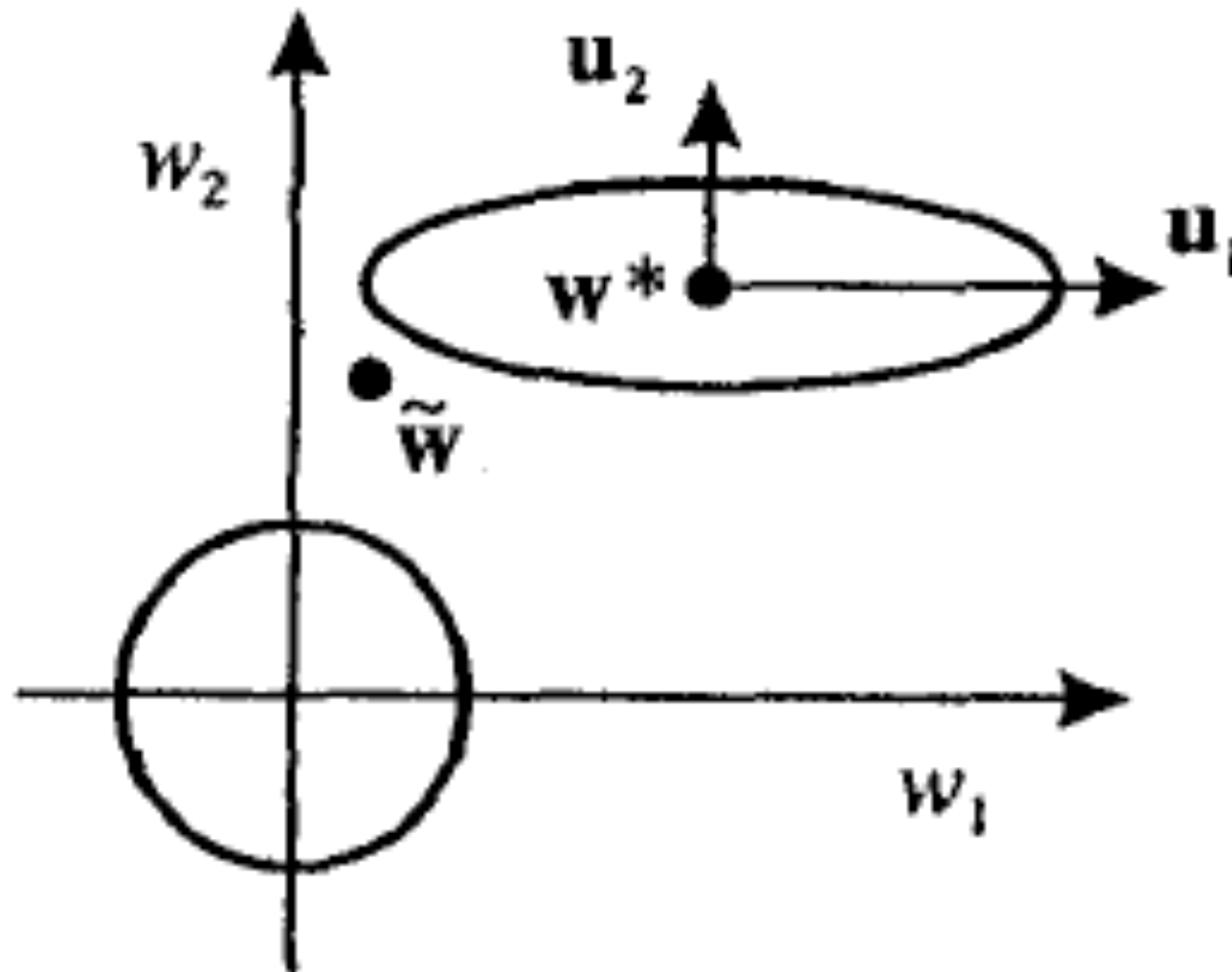
Underfit



Overfit

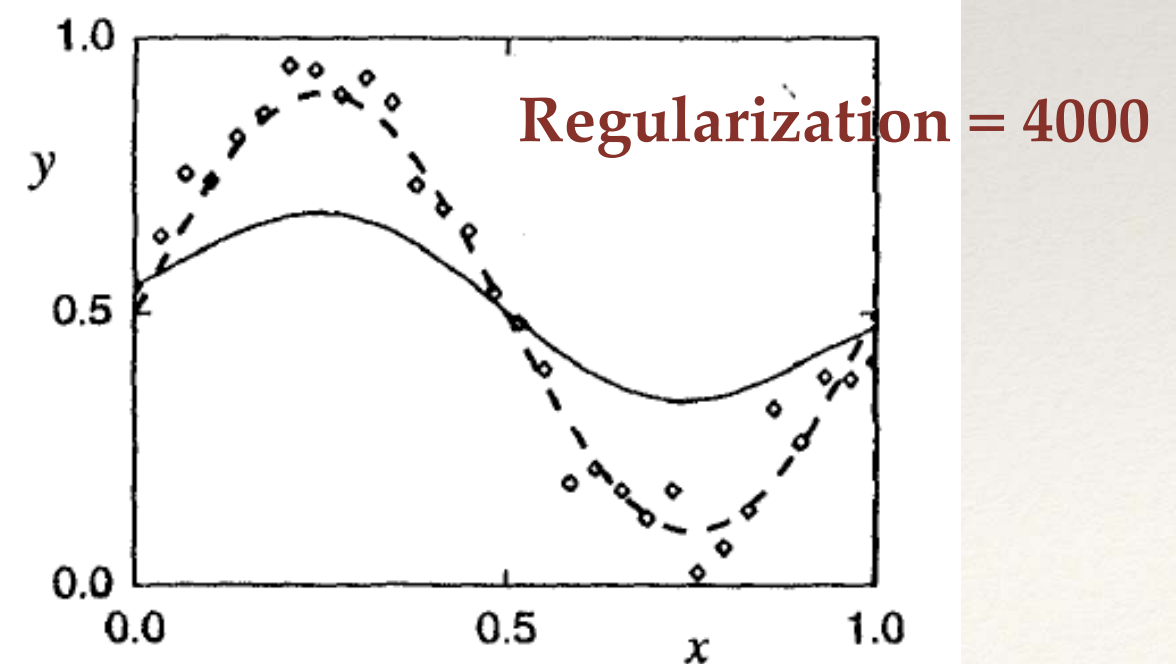
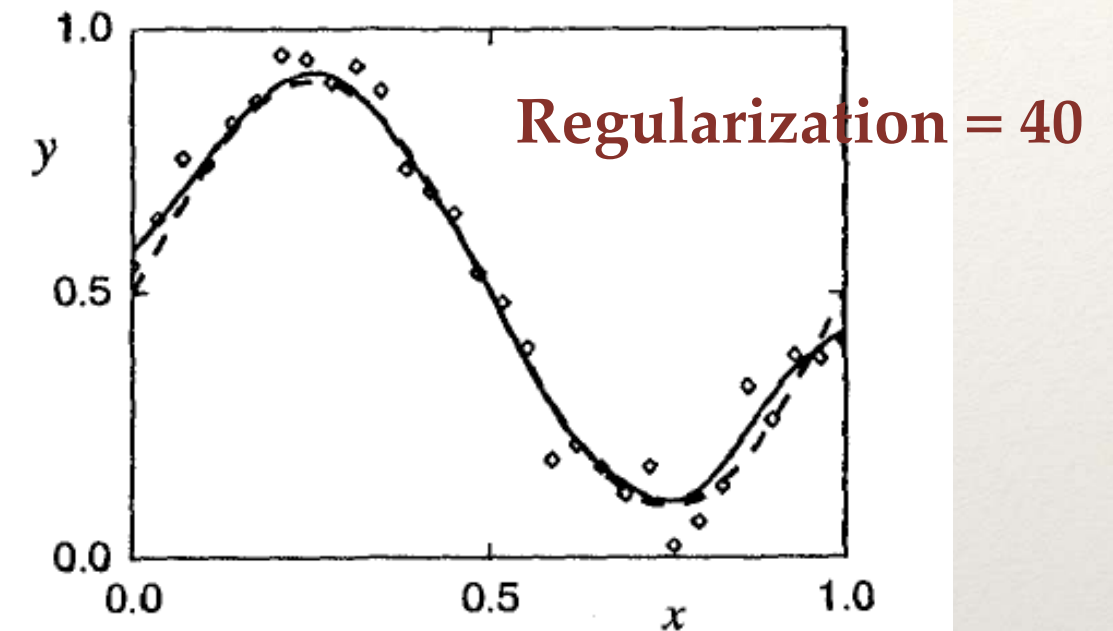
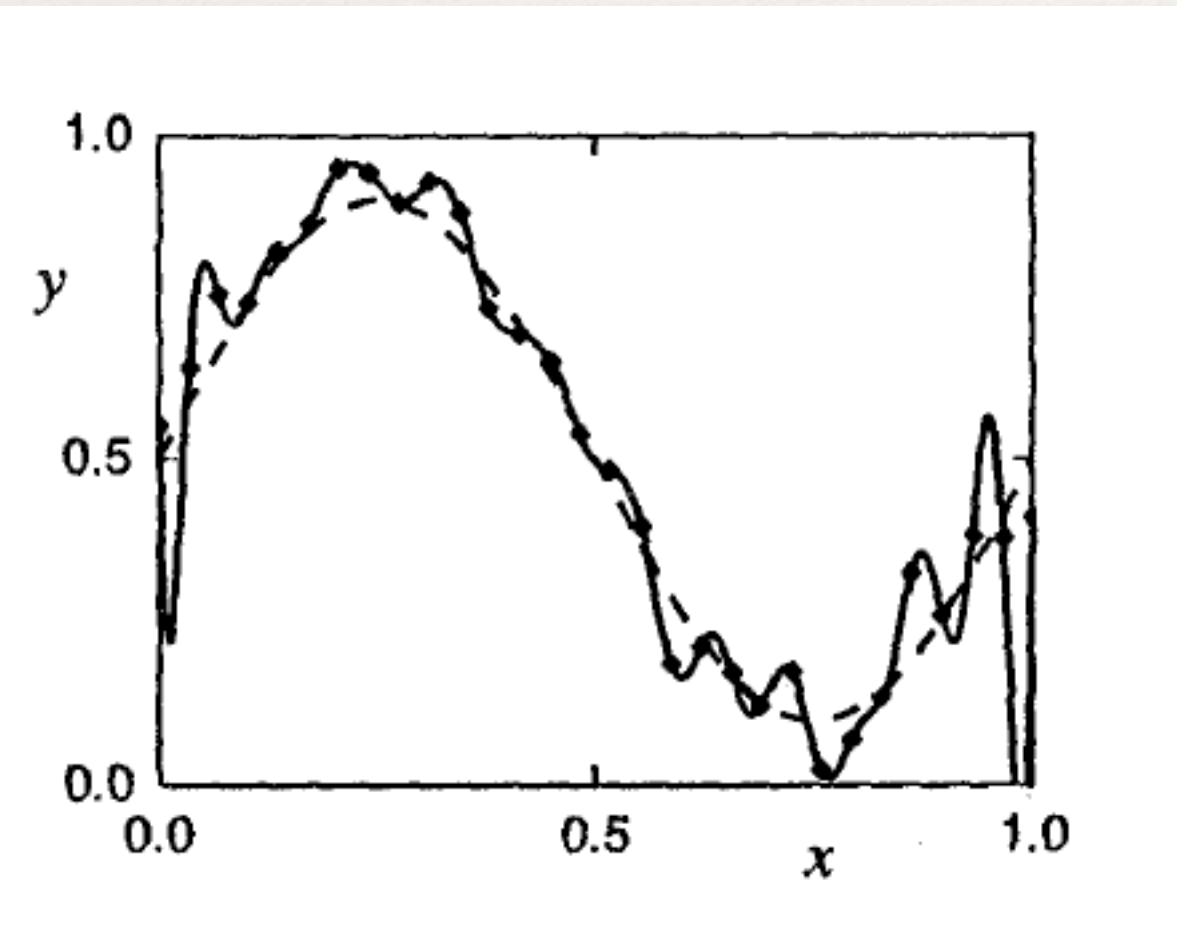


Weight Decay Based Regularization

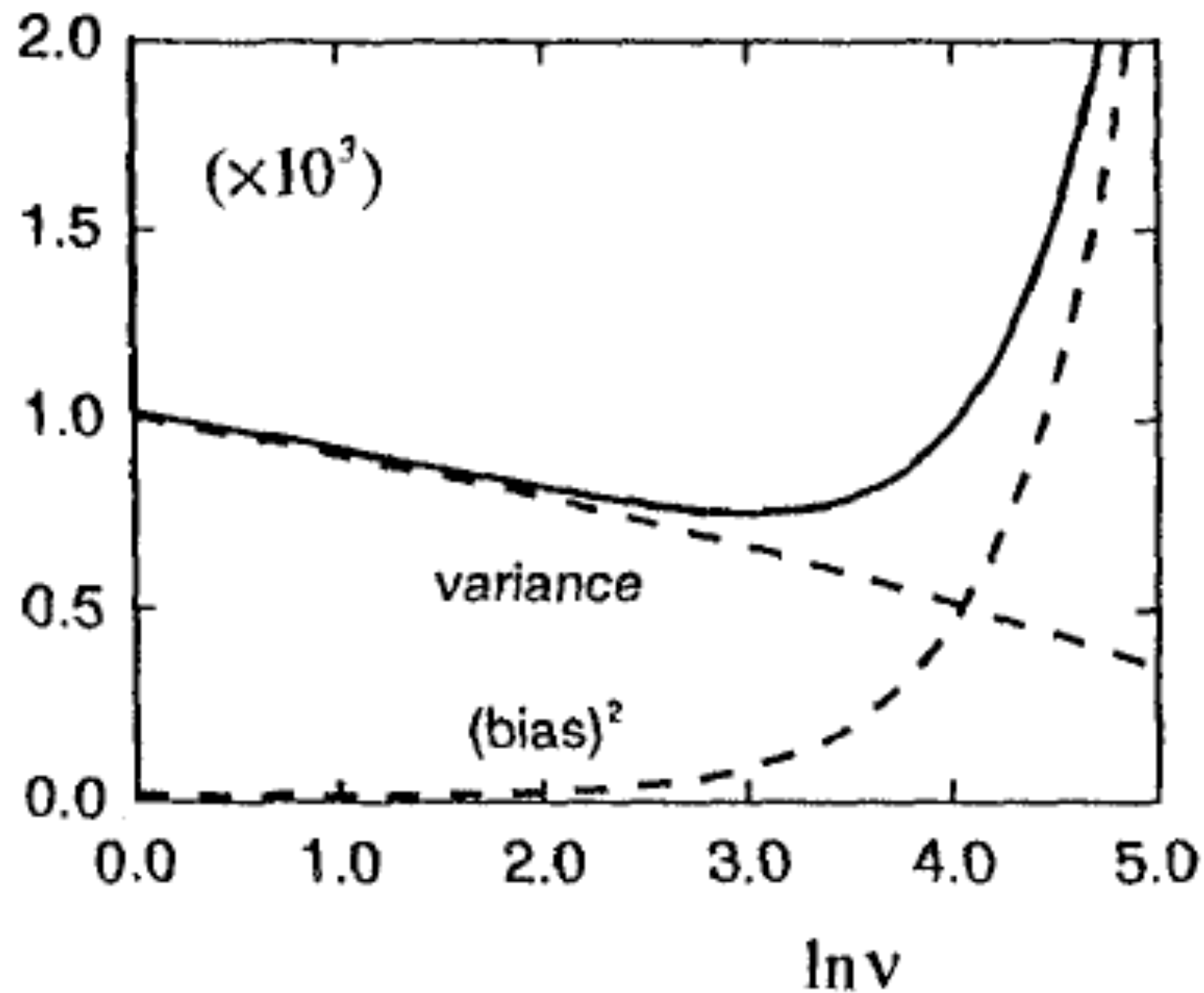


Weight Decay Regularization

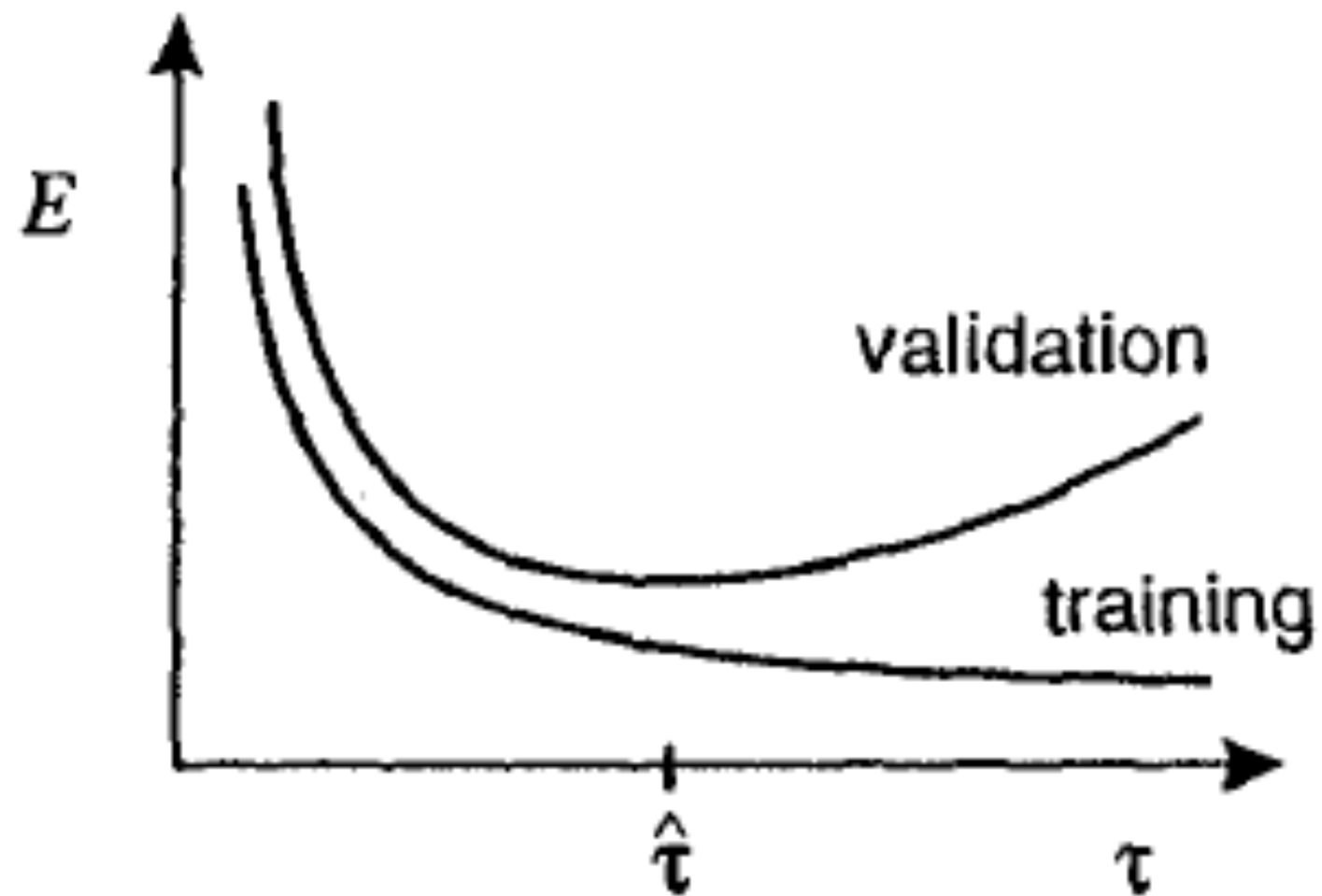
Regularization = 0



Regularization Effect on Learning



Early Stopping



Most Popular in Practice

Neural Networks - Summary

- ❖ Details of Architecture
- ❖ Computation of gradient using back propagation.
- ❖ Error function and output layer activation
 - ❖ Neural networks estimate posterior probabilities
- ❖ Learning in Neural networks
 - ❖ Gradient descent - Properties
- ❖ Generalization of Neural Networks

Batch Normalization

Batch Normalization: Accelerating Deep Network Training by
Reducing Internal Covariate Shift

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Effect of Batch Normalization

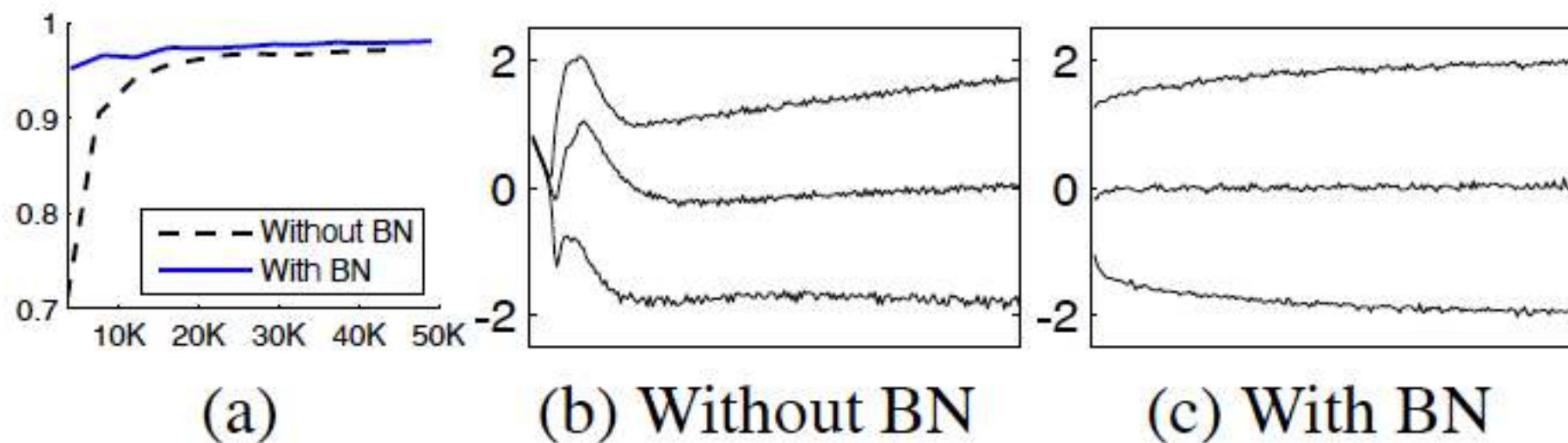


Figure 1: (a) *The test accuracy of the MNIST network trained with and without Batch Normalization, vs. the number of training steps. Batch Normalization helps the network train faster and achieve higher accuracy.* (b, c) *The evolution of input distributions to a typical sigmoid, over the course of training, shown as {15, 50, 85}th percentiles. Batch Normalization makes the distribution more stable and reduces the internal covariate shift.*

Dropout Strategy in Neural Network Training

Dropout: A Simple Way to Prevent Neural Networks from Overfitting

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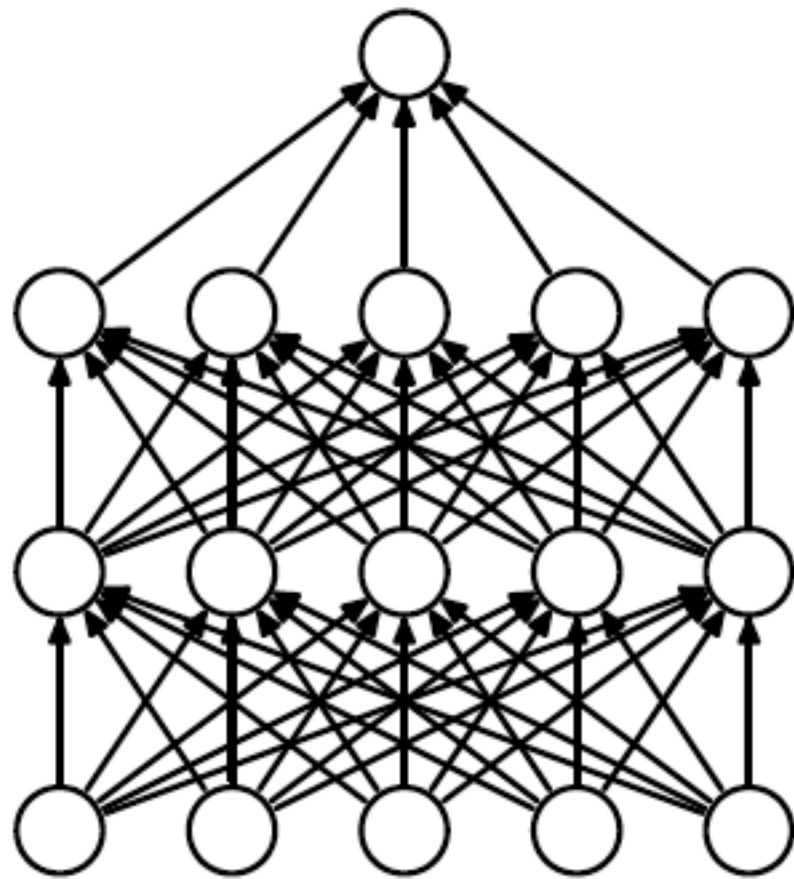
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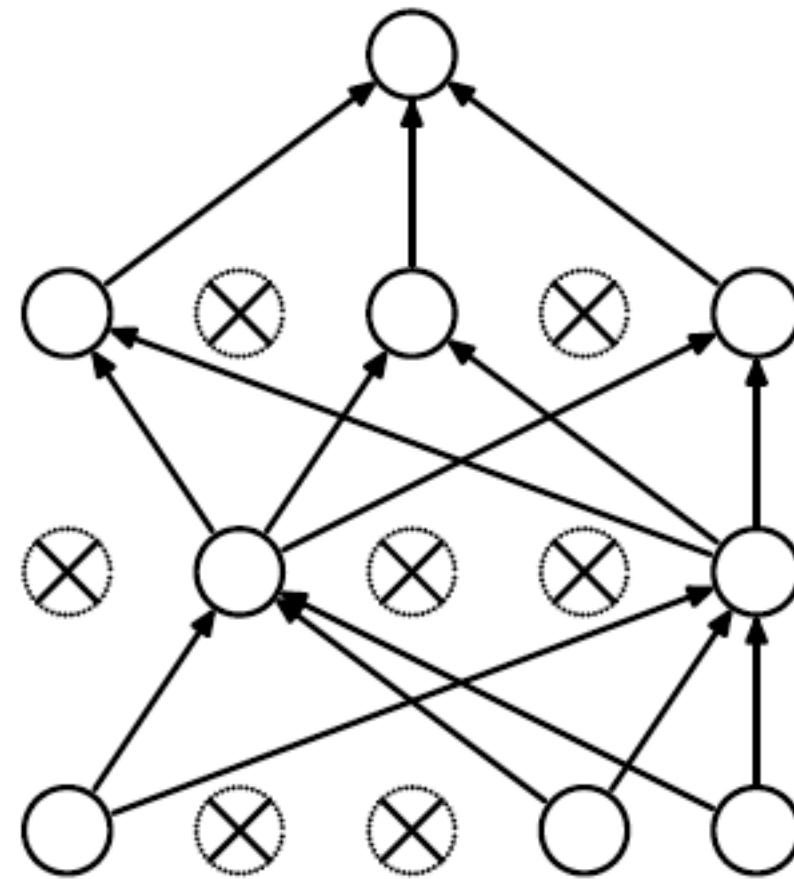
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Editor: Yoshua Bengio

Dropouts in Neural Networks

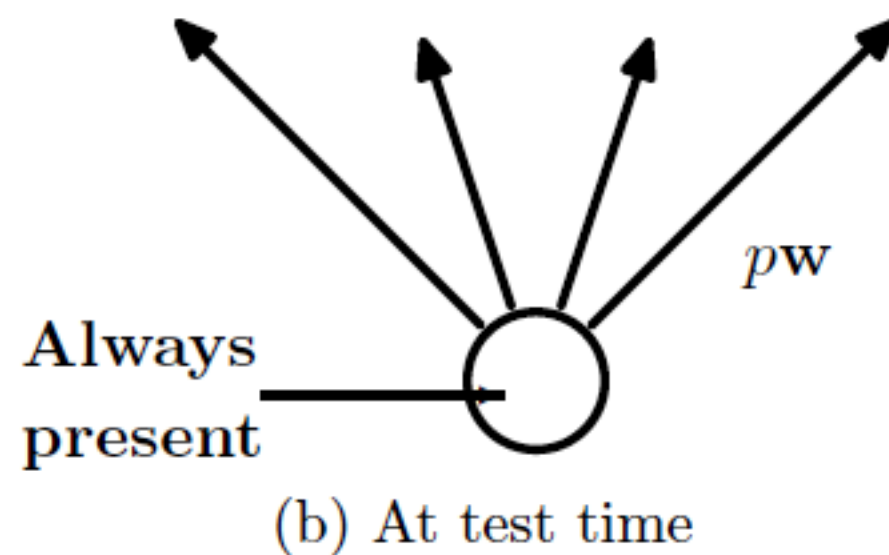
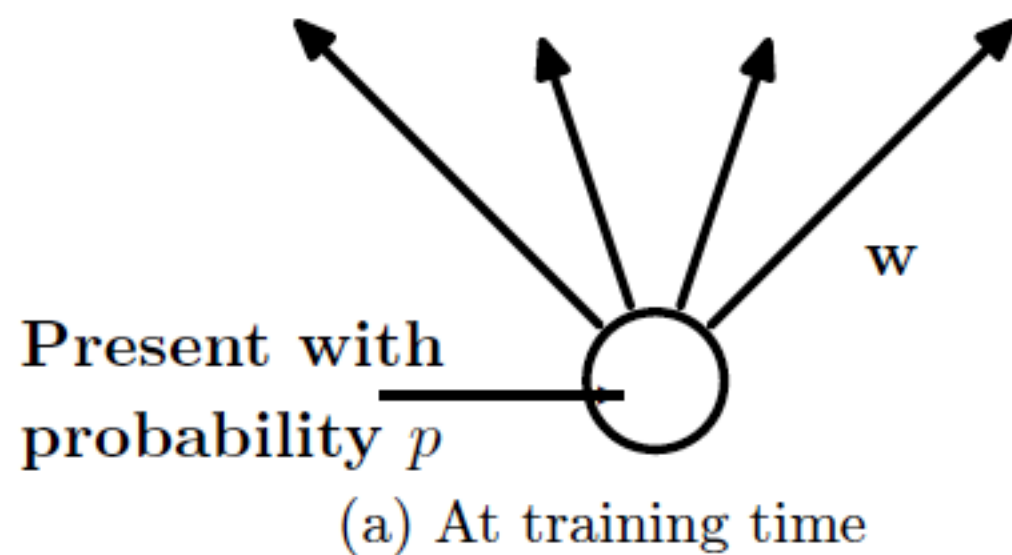


(a) Standard Neural Net

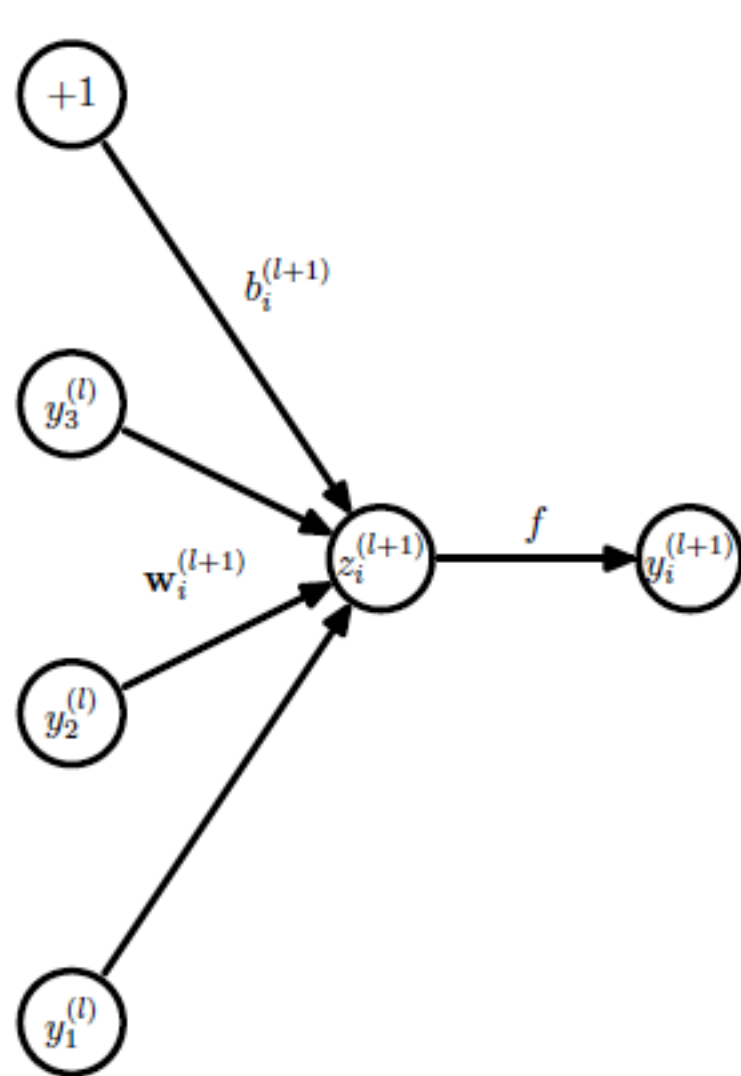


(b) After applying dropout.

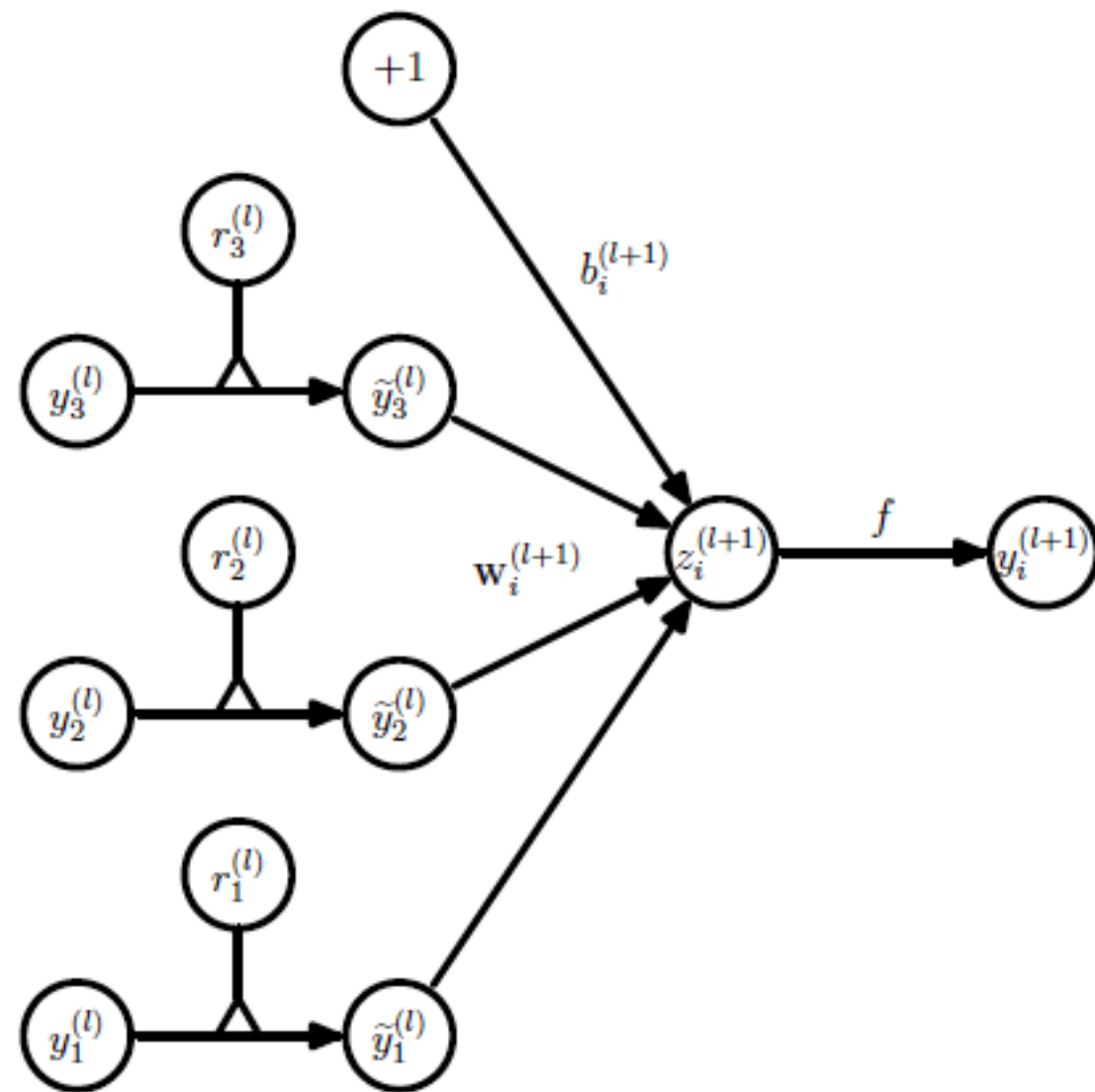
Dropout in Training and Test



Dropout Application



(a) Standard network



(b) Dropout network

Figure 3: Comparison of the basic operations of a standard and dropout network.

Effect of Dropouts

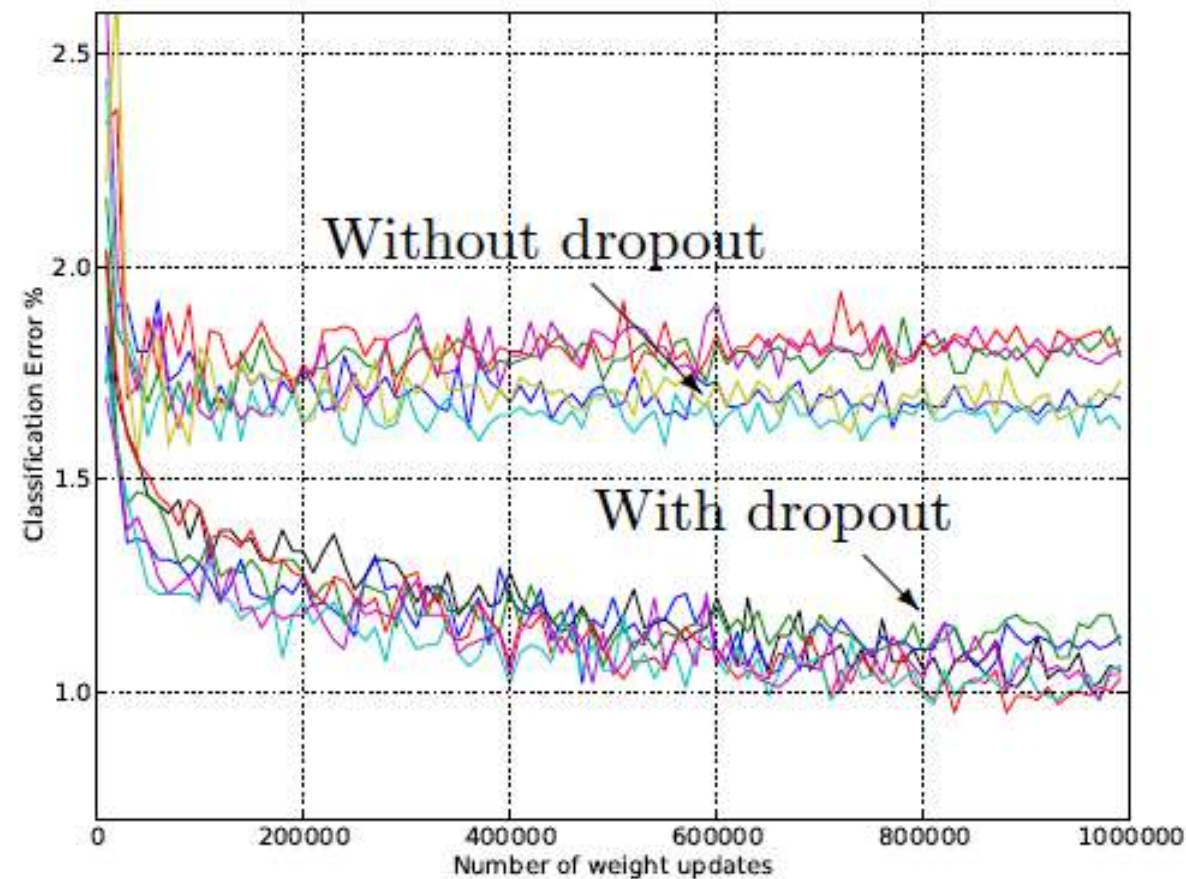


Figure 4: Test error for different architectures with and without dropout. The networks have 2 to 4 hidden layers each with 1024 to 2048 units.